



The internet is broken. And to fix it, researchers want to start all over again

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It's a boring Sunday, and you've decided to while away the time in a wonderfully unproductive game of Tetris. At some point, you're accidentally (or purposefully) going to plug an empty space, temporarily throwing away your chances at scoring a line. "It's all right," you think, "I'll just finish these lines on top first, and then I can get to that later". But inevitably, you end up plugging another vital space, and then another. Pretty soon, the bricks have mounted high enough for you to start worrying. Today, the internet is that game of Tetris, and the bricks are stacked uncomfortably high.

Game over?

This might sound familiar: "The internet is running out of IP addresses!". Even in the early days of the internet, scientists were beginning to realise that this system was getting too popular too fast — they predicted that the world would run out of IP addresses by 1994. Of course, that didn't happen. The internet had plenty of addresses to spare in 1994, and continues to chug along happily even today. This is just another hole we're going to come back to later. Meanwhile, we're finishing the other rows — like security.

Hang on

The internet *isn't* secure. While you're hitting the Report Spam button on your email account, someone's watching an adware program spew out its horrifying results, and some other poor soul can't tell that his PC is part of a botnet, helping send you the spam you're reporting. Firewalls are nice, but the fact that you're still receiving spam and fighting off phishing attacks speaks for itself. And the root of all these problems (predictably enough) is stupidity.

The dumb network

In networking-speak, a *dumb network* is essentially a bunch of terminals and wires. It doesn't care about the information travelling on it — just that it's taking the information where it needs to go. The terminals themselves — the computers — are pretty smart, so the onus was on them to interpret the data and present it to users.

There is much to love about the dumb network — mostly, it's economical. A dumb network is ridiculously easy (and hence cheap), to deploy — a router here, a cable there, and one more happy customer gets connected to the web. It's also extensible — because a dumb network is like a clean slate, you can add more functionality to it as you go along: perfect for a network that grew as fast as the internet did. And finally, by putting the onus of processing data on the terminals, you're creating a network that lets data travel quickly, without obstructions.

The internet is a *packet-switched* dumb network, which means that the information you send along the pipes is broken up into *packets*, sent along different paths to the destination, and then re-assembled at the other end. This is all very well for data; send voice over a line like that, however, and you get gibberish at the

other end — if anything. It took ten years for VoIP to work around this mess; but even now, it's too unreliable to replace the good old wired phone line.

Still, VoIP gets the job done. Ish. The *real* problem with the dumb network is that it isn't secure.

The trusting network

The internet was designed by a bunch of scientists, for the scientific community. It was a good-natured network that had faith in its users, and wouldn't believe that people would actually send malicious data to each other. Then it got misused — malicious people used it to spread malicious code, and the internet suffered a bad reputation. Now, of course, it's become somewhat smarter. By installing firewalls and packet sniffers (software that checks packets for malicious data) on servers, routers and your PCs, the internet has made it much harder for evil code to get to you, unless you're conned into letting it in yourself.

But there still remains one part of the net that's still good-natured and trusting at its core — DNS. It was originally designed to be the digital equivalent of the friendly guy at the concierge's desk. You would ask it for a web address, and it would helpfully tell you where to go. Then came *cache poisoning*, by which hackers could fool DNS servers into believing they're sending you to the fun part of town, but you really end up in a dark alley with your head bashed in. And somewhere underneath all the security protocols, this flaw still exists.

It's time to re-do the internet, and turn it into a smart, secure network.

What we need

The new internet needs to be faster and more reliable — IP telephony shouldn't suffer the way it always has, and must eventually become the alternative to the classic telephone line that it's supposed to be.

And yes, it would also be nice if the new internet could let our devices have their own unique IP addresses, instead of an internal address that a router or server assigns to it. We'd also like the internet to be "ready" for when we want to use our mobile phones as our primary computing devices.

However, before researchers are to give us the internet we desire, they need a sandbox to test their designs — a guinea-pig network.

The playground

In 2005, the National Science Foundation (NSF) in the US asked researchers to come up with a new plan for the internet. To help them test their designs, the NSF started the Global Environment for Networking Innovation (GENI), which would be the guinea-pig internet for them to use. Several supporting projects have spawned since, each with their own view of what the internet should be.

It isn't hard to guess which side of the fence Stanford's Clean Slate Design project is on. The group is creating a set of technologies that they hope will form the backbone of the new internet — from a framework for a mobile internet, to an infrastructure for